

Special Issue on Recent Advances in Machine Learning and Computational Intelligence

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1. Introduction

Machine learning and computational intelligence are currently high-profile research areas attracting the attention of many researchers. They have achieved remarkable results in various fields such as computer vision and natural language processing, showing their strong advantages. Researchers have explored many intelligent algorithms, characterized by computational adaptability, robustness, and high performance. These algorithms facilitate intelligent behavior in complex and dynamic environments and the development of technology that enables machines to think, behave, or act more humanely. This not only promotes the further development of machine learning and computational intelligence, but also provides richer ideas for applications.

2. Recent Advances

In view of the above, this Special Issue was introduced to collect the latest research on this topic. These latest research works have addressed various practical application scenarios by utilizing machine learning and computational intelligence techniques. This Special Issue contains 10 papers written by research experts on related topics of interest. In reviewing this Special Issue, various topics have been addressed, predominantly machine learning techniques and heuristic search algorithms. The following seven papers utilize machine learning techniques to solve problems in the fields of computer vision, natural language processing, classification, and so on. The first paper, authored by M. Kim and M.H. Song, studies the machine learning-based diagnosis of facial skin problems. They used enhanced mask R-CNN and super-resolution GAN to successfully solve this problem [1]. The second paper was written by Z. Liu, X. He and Y. Lu. This paper addresses the problem of left ventricle (LV) segmentation of cardiac magnetic resonance (MR) images, which could help doctors in the clinical diagnosis of cardiovascular diseases (CVDs). They provided an effective solution by combining the strengths of UNet 3+ and Transformer [2]. The authors A.M. Mostafa, M. Aljasir, M. Alruily, A. Alsayat, and M. Ezz provided a comprehensive review of recent sentiment analysis methods based on lexicon or machine learning. They proposed a forward fusion feature selection algorithm for the sentiment analysis problem of Arabic reviews [3]. The fourth paper, by G. Ou, Y. He, P. Fournier-Viger and J.Z. Huang, proposed a new naive Bayesian classifier (NBC) construction method for mixed attribute data classification problems [4]. This method is mainly intended to solve two limitations of the NBC: one is the assumption of strong independence; while the other is that it cannot effectively solve continuous attributes. The fifth paper, by Z. Zhang, X. Chang, H. Ma, H. An, and L. Lang, proposed a new locomotion control algorithm for quadruped robots by combining the advantages of model predictive control (MPC) and reinforcement learning (RL) [5]. It is an adaptive approach that achieves a better locomotion performance and balance stability. The sixth paper, by C. Wang and Z. Xiao, used deep



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learning methods to study credit scoring problems in the financial industry, and designed an end-to-end feature-embedded transformer (FE-Transformer) credit scoring method [6]. The seventh paper, authored by Y. Huang, X. Xu, Y. Li, X. Zhang, Y. Liu, and X. Zhang, used deep reinforcement learning techniques to solve the vehicle-following control problem [7]. They proposed a subsection proximal policy optimization method (subsection-PPO) to improve the efficiency and safety of the vehicle-following control method.

In addition, three papers studied heuristic search algorithms in the field of computational intelligence. In the first paper, L. Zhao and H. Jin improved the traditional vector-weighted optimization algorithm (INFO) and designed a promising optimization algorithm (IDEINFO) [8]. The algorithm further improves the global search ability and achieves an excellent optimization performance. The second paper showed a new UAV path planning algorithm (RJA-Star), proposed by J. Li, W. Zhang, Y. Hu, S. Fu, C. Liao, and W. Yu [9]. This method significantly reduced the moving distance, computation time, number of nodes, number of corners, and maximum angles, and effectively improved the obstacle avoidance ability of agricultural drones. The final paper, by A. Aboud, N. Rokbani, B. Neji, Z. Al Barakeh, S. Mirjalili, and A.M. Alimi, studied the use of the crow search algorithm (CSA) for dynamic multi-objective optimization and multi-objective optimization problems [10]. The authors designed a distributed bi-behaviors crow search algorithm (DB-CSA) with two new mechanisms.

3. Future Outlook

This Special Issue introduces many novel and interesting methods, providing guidance for the further development of machine learning and computational intelligence. Looking forward, there are still many thought-provoking issues worthy of further in-depth exploration by researchers. In the future, it will be necessary to apply machine learning and computational intelligence technologies to solve more challenging problems in various fields and propose more robust, accurate, and efficient solutions.

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